MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC
RECTIFIER, LOW CURRENT, POWER, KEY CODE GPR

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic low current rectifier module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Rectifier, low current
Family	Power
Key code	GPR
Size	1A
Weight	25 grams maximum
Failure rate	0.92 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	3 W
Rectifier forward voltage drop	
If = 0.5 mA	0.23 V minimum, 0.60 V maximum
If = 100 mA	0.41 V minimum, 0.93 V maximum
If = 1.75 A	0.63 V minimum, 1.38 V maximum
Rectifier inverse current	
Vr = 10 V	5.0 µA maximum
Vr = 150 V	20 µA maximum
Vr = 400 V	100 µA maximum
Rectifier reverse recovery time	
If = Ir = 0.6 A	350 ns maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D. C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Maximum required limits	
Rectifier continuous forward current	1.75 A
Rectifier peak inverse voltage	400 V
Rectifier peak one cycle (60 Hz) surge	35 A

1.3 Module function. This module contains 12 independent 1.75 A diodes.

1.3.1 Forward current. When one rectifier per module is operated, the continuous forward current shall not exceed 1.75 A, however, any other combinations of rectifiers may be operated provided the 3 W maximum module power dissipation requirement is not exceeded.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.92 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I, as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GPR.

3.5.2 Size. The module size shall be 1A.

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3.5.3 Weight. The module shall have a maximum weight of 25 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/24, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

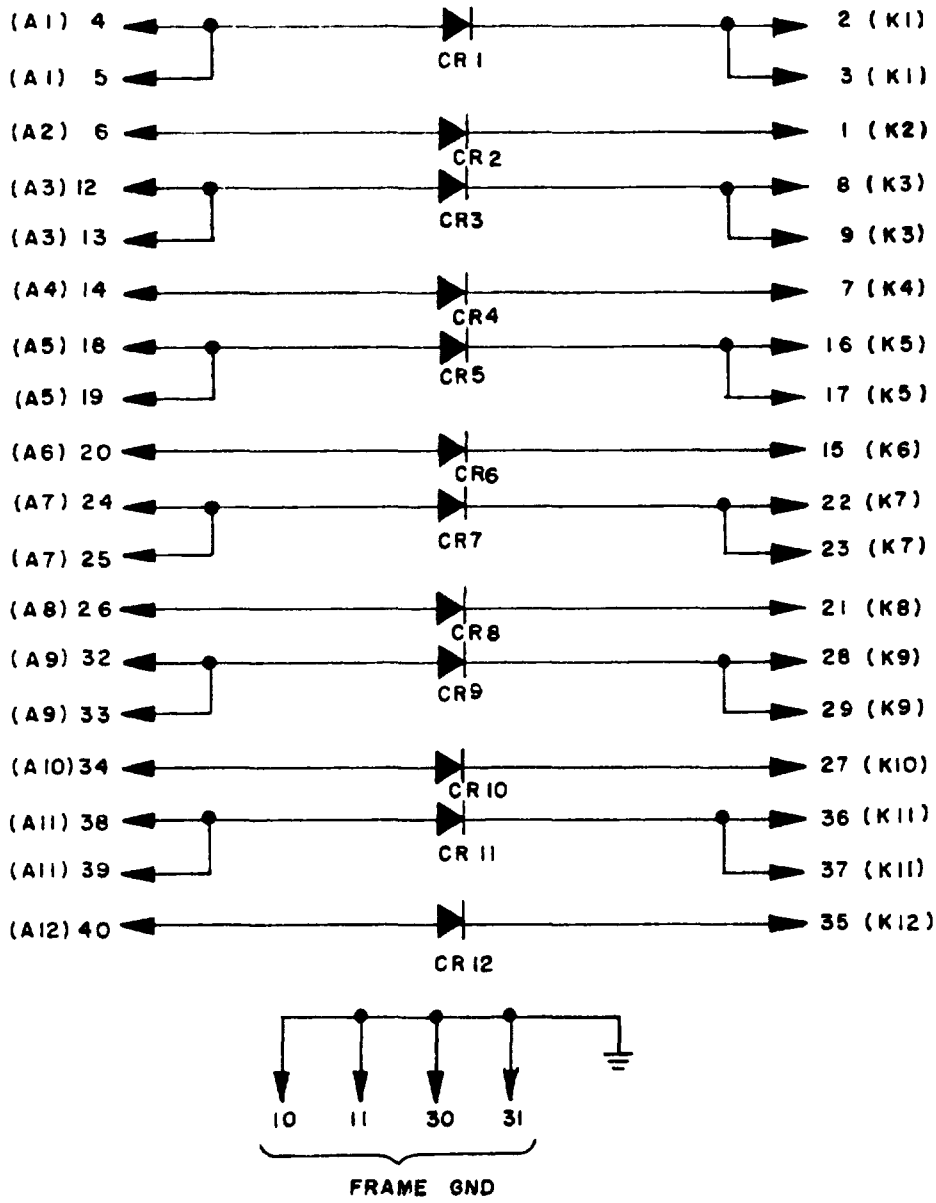
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment		MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply		Item P01	1
DC voltmeter		Item V04	1
DC ammeter		Item X01	1
DC power supply		Item P03	1
Oscilloscope		Item D01	1
Oscilloscope probe		Item Y01	1
Pulse generator		Item J04	1
Capacitance meter		Item C05	1
Megohm bridge		Item R06	1
Ohmmeter		Item R10	1
Resistors			
23.7Ω ±1%	3 W		1
1.0Ω ±1%	4 W		1
100Ω ±1%	2 W		1
2 kΩ ±1%	1/2 W		1
10 kΩ ±1%	1/2 W		1
30Ω ±5%	50 W, noninductive		1
1.0Ω ±1%	1 W, noninductive		1
11Ω ±10%	1 W, noninductive		1
100Ω ±10%	1 W, noninductive		1

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NOTE:
A1-12 = ANODE
K1-12 = CATHODE

FIGURE 1. GPR module function diagram, low current rectifier.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	Forward voltage drop (If = 100 mA)	II	X	0.55	0.81	0.50	0.86	0.46	0.88	0.41	0.93	V
B1-B18	Forward voltage drop (If = 0.5 mA)	III	X	0.35	0.50	0.30	0.55	0.27	0.55	0.23	0.60	V
C1-C12	Forward voltage drop (If = 1.75 A)	IV	S	0.75	1.28	0.70	1.33	0.68	1.32	0.63	1.38	V
D1-D12	Inverse current (Vr = 400 V)	V	S	0	10	0	10	0	100.0	0	100.0	µA
E1-E12	Inverse current (Vr = 150 V)	VI	X	0	3	0	3	0	20	0	20	µA
F1-F12	Inverse current (Vr = 10 V)	VII	S	0	1	0	1	0	5	0	5	µA
G1-G12	Reverse recovery time	VIII	S	0	350	0	350	0	350	0	350	ns
H1-H12	Junction capacitance	IX	S	0	240	0	240	0	240	0	240	pF
J1-J7	Insulation resistance	X	X	10	-	1	-	10	-	1	-	MΩ
K1-K4	Continuity	XI	S	0	1.0	0	1.0	--	--	--	--	Ω
L1-L12	Peak one cycle surge current	XII	Q	35	--	--	--	--	--	--	--	A

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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Equipment	No. required (maximum per test figure
Capacitors	
500 µF, - 10%, + 100%, 50 Vdc, electrolytic	1
Coil	
1.0 ±0.5 µH	1
Switch	
Single pole, momentary, normally open	1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through L shall be performed in any order with the following exceptions. If test conditions D, J, and L are part of a sequence of tests, the order of testing shall be as follows L, D, J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check test shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A, B, E, and J shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown in figure 2.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through L shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

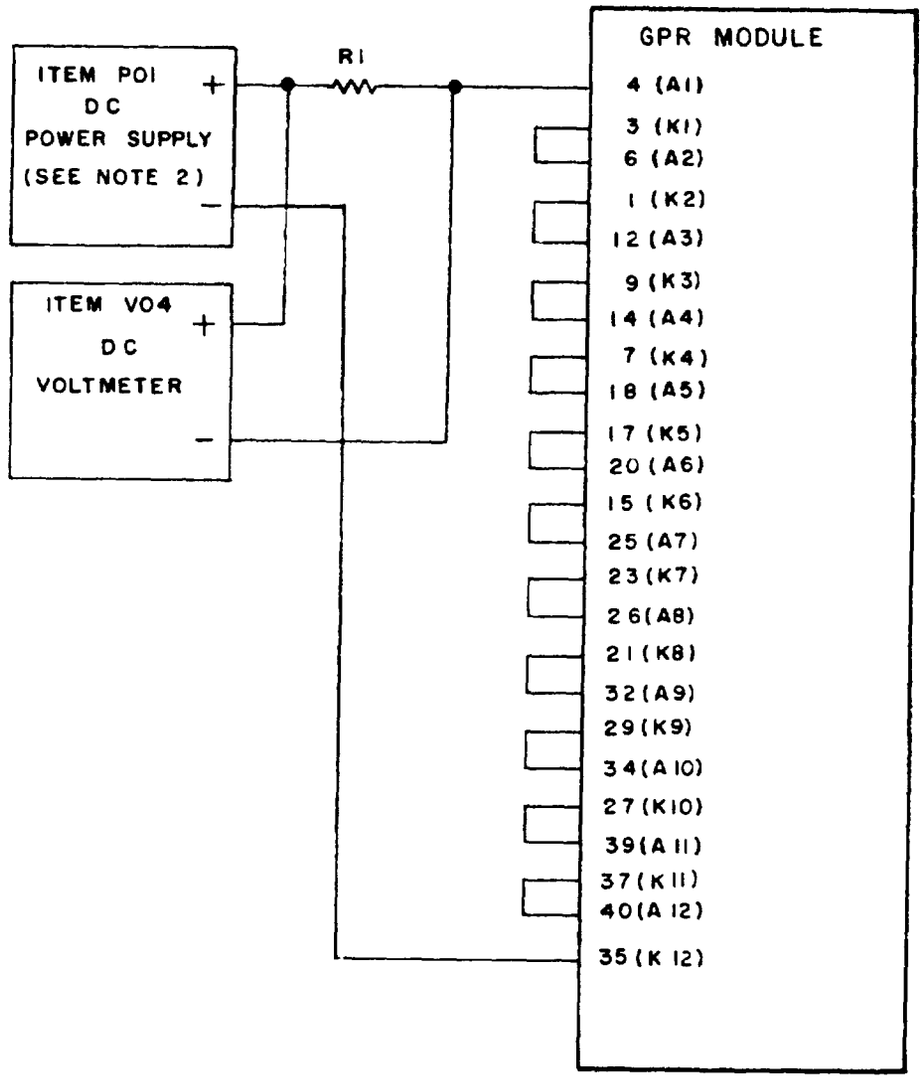
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

Preparing activity.
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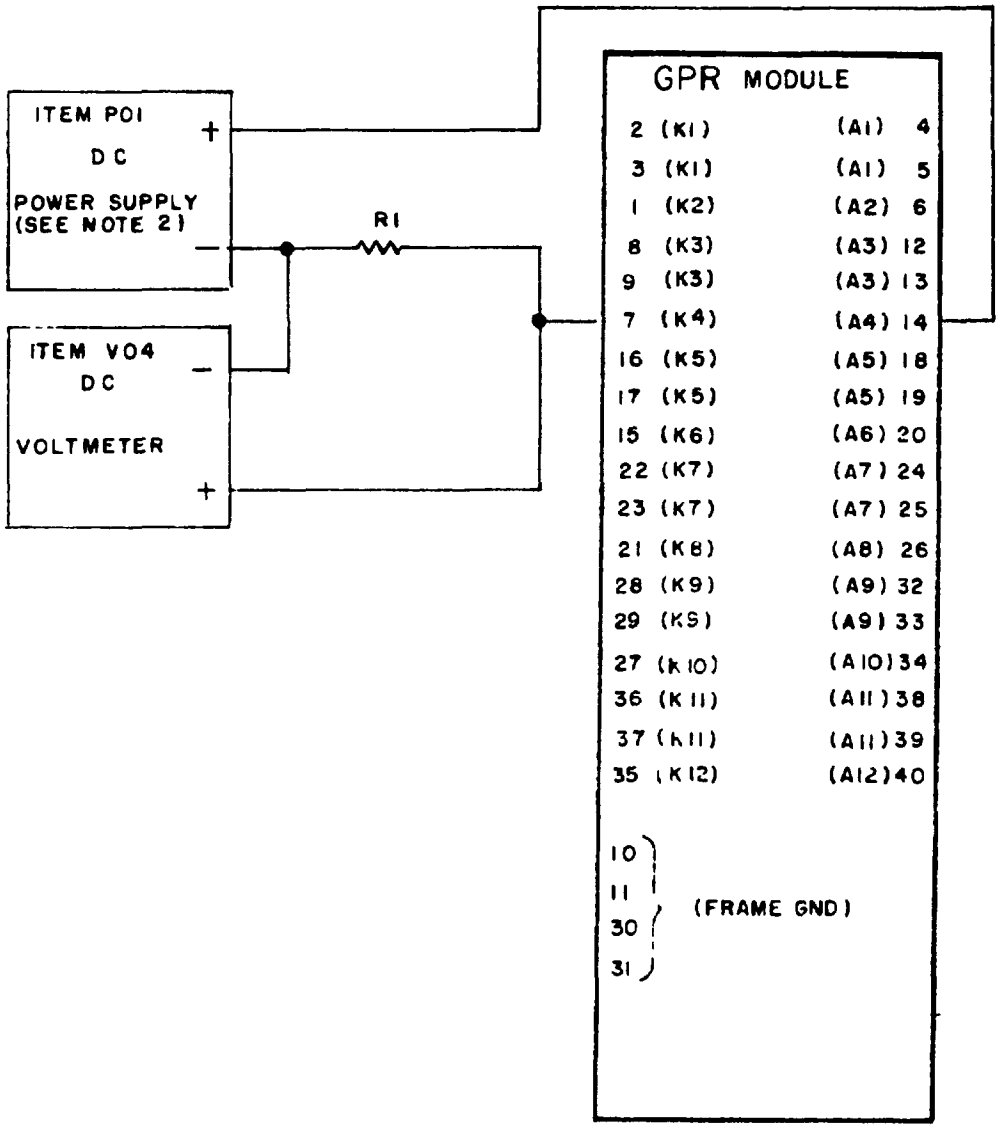
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NOTES:
1. R1 = 23.7Ω ±1%, 3 W.
2 Adjust power supply P01 for a voltage reading on voltmeter V04 of 7.11 V ±1%, corresponding to a current of 300 mA.

FIGURE 2. Operating temperature transient thermal, vibration, and life test circuit.

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NOTES:
1. R1 = 1.0Ω ±1%, 4 W.
2. Adjust power supply P01 for a voltage reading on voltmeter V04 of 1.75 V ±1%, corresponding to a current of 1.75 A.
3 Reconnect voltmeter V04 (+ to pin 14, - to pin 7) to check forward voltage drop for 1 38 V, maximum.

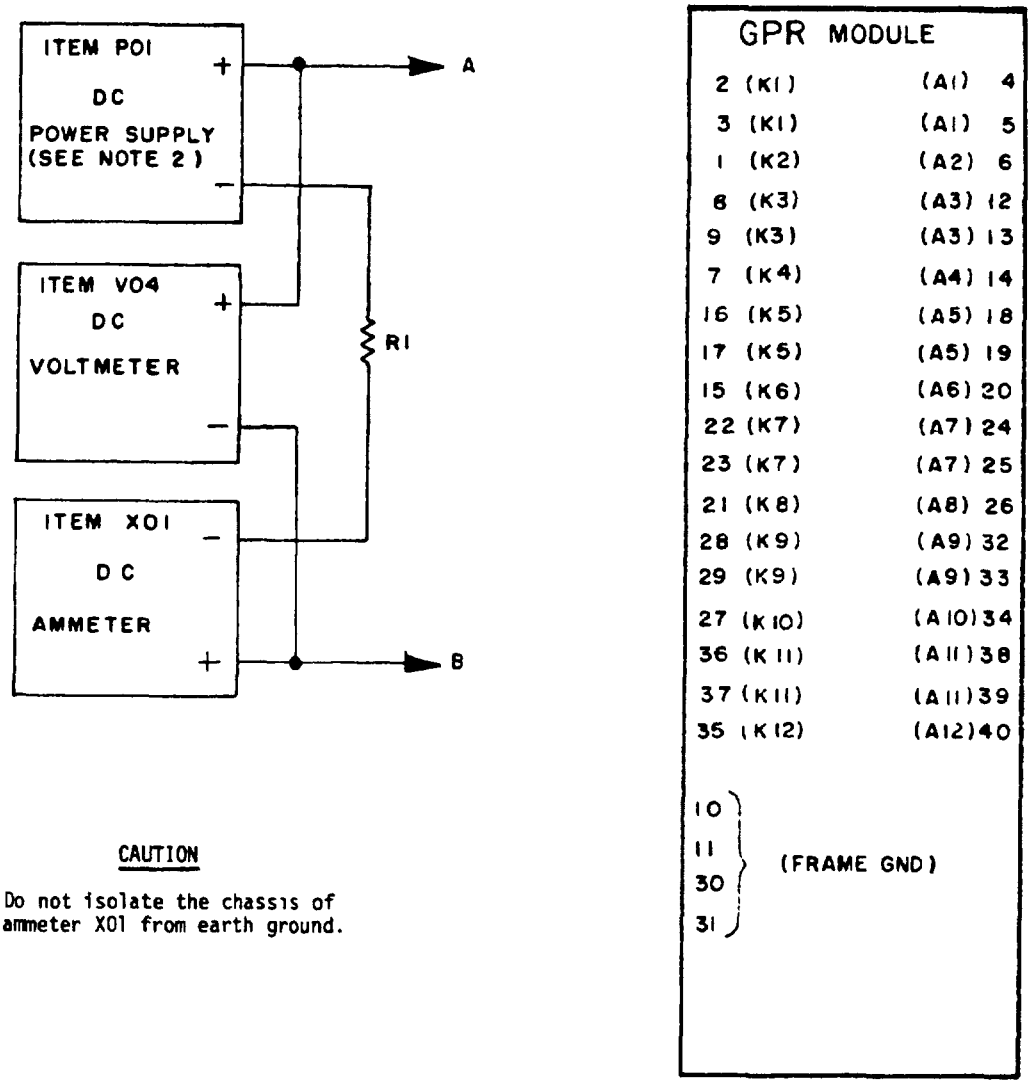
FIGURE 3. Critical component temperature test circuit.

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TABLE II. Forward voltage drop (If=100 mA) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Forward voltage drop (If=100 mA)	4	3	0.55	0.81	V	4
A2		6	1				
A3		12	9				
A4		14	7				
A5		18	17				
A6		20	15				
A7		25	23				
A8		26	21				
A9		32	29				
A10		34	27				
A11		39	36				
A12	Forward voltage drop (If=100 mA)	40	35	0.55	0.81	V	4

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 100Ω ±1%, 2 W
2. Connect points A and B to the module and adjust power supply P01 for a current of 100 mA ±1% as indicated on ammeter X01.
3. Read the voltage drop on voltmeter V04.

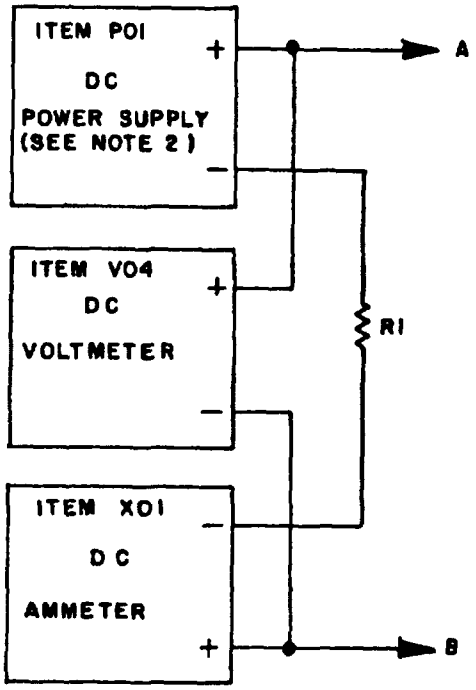
FIGURE 4. Diode forward voltage drop (If=100 mA) test circuit for table II.

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TABLE III. Forward voltage drop (If=0.5 mA) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Forward voltage drop (If=0.5 mA)	4	2	0.35	0.50	V	5
B2		5	2				
B3		6	1				
B4		12	8				
B5		13	8				
B6		14	7				
B7		18	16				
B8		19	16				
B9		20	15				
B10		24	22				
B11		25	22				
B12		26	21				
B13		32	28				
B14		33	28				
B15		34	27				
B16		38	37				
B17		39	37				
B18	Forward voltage drop (If=0.5 mA)	40	35	0.35	0.50	V	5

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES
1. R1 = 2kΩ ±1%, 1/2 W
2. Connect points A and B to module and adjust power supply PO1 for a current of 0.5 mA ±1% as indicated on ammeter XO1.
3 Read the voltage drop on voltmeter VO4.

GPR MODULE	
2 (K1)	(A1) 4
3 (K1)	(A1) 5
1 (K2)	(A2) 6
8 (K3)	(A3) 12
9 (K3)	(A3) 13
7 (K4)	(A4) 14
16 (K5)	(A5) 18
17 (K5)	(A5) 19
15 (K6)	(A6) 20
22 (K7)	(A7) 24
23 (K7)	(A7) 25
21 (K8)	(A8) 26
28 (K9)	(A9) 32
29 (K9)	(A9) 33
27 (K10)	(A10) 34
36 (K11)	(A11) 38
37 (K11)	(A11) 39
35 (K12)	(A12) 40
10, 11, 30, 31	(FRAME GND)

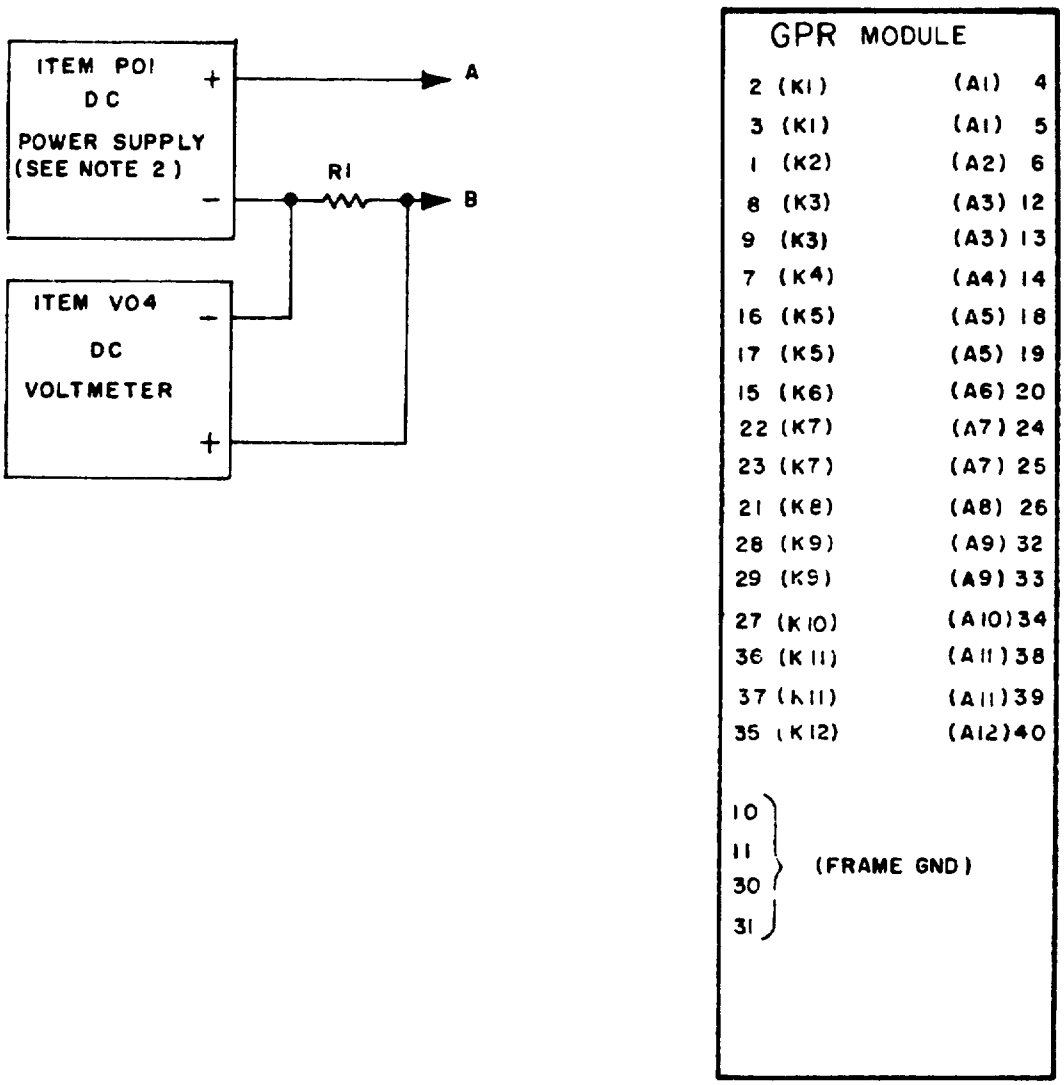
FIGURE 5. Diode forward voltage drop (If=0.5 mA) test circuit for table III.

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TABLE IV. Forward voltage drop (If=1.75 A) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Forward voltage drop (If=1.75 A)	5	3	0.75	1.28	V	6
C2		6	1				
C3		13	9				
C4		14	7				
C5		19	17				
C6		20	15				
C7		24	23				
C8		26	21				
C9		33	29				
C10		34	27				
C11		39	37				
C12	Forward voltage drop (If=1.75 A)	40	35	0.75	1.28	V	6

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NOTES:
1. R1 = 1.0Ω ±1%, 4 W.
2. Connect points A and B to module and adjust power supply P01 for a voltage reading on voltmeter V04 of 1.75 V ±1%.
3. Reconnect voltmeter V04 (+ to A, - to B) to measure the forward voltage drop.

FIGURE 6. Diode forward voltage drop (If=1.75 A) test circuit for table IV.

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TABLE V. Inverse current (Vr=400 V) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Inverse current (Vr=400 V)	4	3	0	10	µA	7
D2		6	1				
D3		12	9				
D4		14	7				
D5		18	17				
D6		20	15				
D7		25	23				
D8		26	21				
D9		32	29				
D10		34	27				
D11		39	37				
D12	Inverse current (Vr=400 V)	40	35	0	10	µA	7

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TABLE VI. Inverse current (Vr=150 V) test.

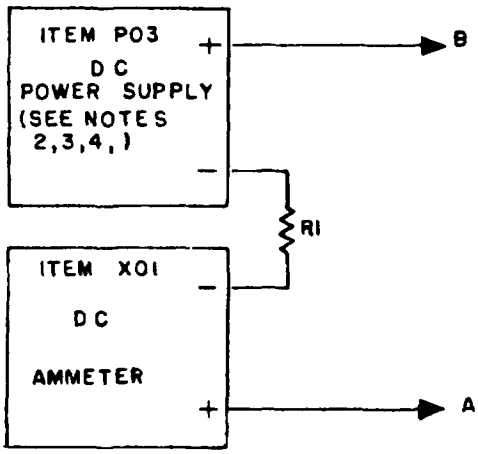
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Inverse current (Vr=150 V)	4	3	0	3	µA	7
E2		6	1				
E3		12	9				
E4		14	7				
E5		18	17				
E6		20	15				
E7		25	23				
E8		26	21				
E9		32	29				
E10		34	27				
E11		39	37				
E12	Inverse current (Vr=150 V)	40	35	0	3	µA	7

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TABLE VII. Inverse current (Vr=10 V) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Inverse current (Vr=10 V)	4	3	0	1	µA	7
F2		6	1				
F3		12	9				
F4		14	7				
F5		18	17				
F6		20	15				
F7		25	23				
F8		26	21				
F9		32	29				
F10		34	27				
F11		39	37				
F12	Inverse current (Vr=10 V)	40	35	0	1	µA	7

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WARNING

High voltage will be present at the output binding posts of item P03 for test conditions D and E Always reduce the output voltage to zero volts before connecting or disconnecting the test leads.

CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

GPR MODULE

2 (K1)	(A1) 4
3 (K1)	(A1) 5
1 (K2)	(A2) 6
8 (K3)	(A3) 12
9 (K3)	(A3) 13
7 (K4)	(A4) 14
16 (K5)	(A5) 18
17 (K5)	(A5) 19
15 (K6)	(A6) 20
22 (K7)	(A7) 24
23 (K7)	(A7) 25
21 (K8)	(A8) 26
28 (K9)	(A9) 32
29 (K9)	(A9) 33
27 (K10)	(A10) 34
36 (K11)	(A11) 38
37 (K11)	(A11) 39
35 (K12)	(A12) 40

10, 11, 30, 31 (FRAME GND)

NOTES:
1. R1 = 10kΩ ±1%, 1/2 W.
2. Adjust the voltage of power supply P03 to 400 +0 V, -20 for table V tests
3. Adjust the voltage of power supply P03 to 150 ±7.5 V for table VI tests.
4. Adjust the voltage of power supply P03 to 10 ±0.5 V for table VII tests.

FIGURE 7. Inverse current (Vr = 400 V, 150 V, 10 V) test circuit for tables V, VI, and VII.

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TABLE VIII. Reverse recovery time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Reverse recovery time	3	4	0	350	ns	8
G2		1	6				
G3		9	12				
G4		7	14				
G5		17	18				
G6		15	20				
G7		23	25				
G8		21	26				
G9		29	32				
G10		27	34				
G11		37	39				
G12	Reverse recovery time	35	40	0	350	ns	8

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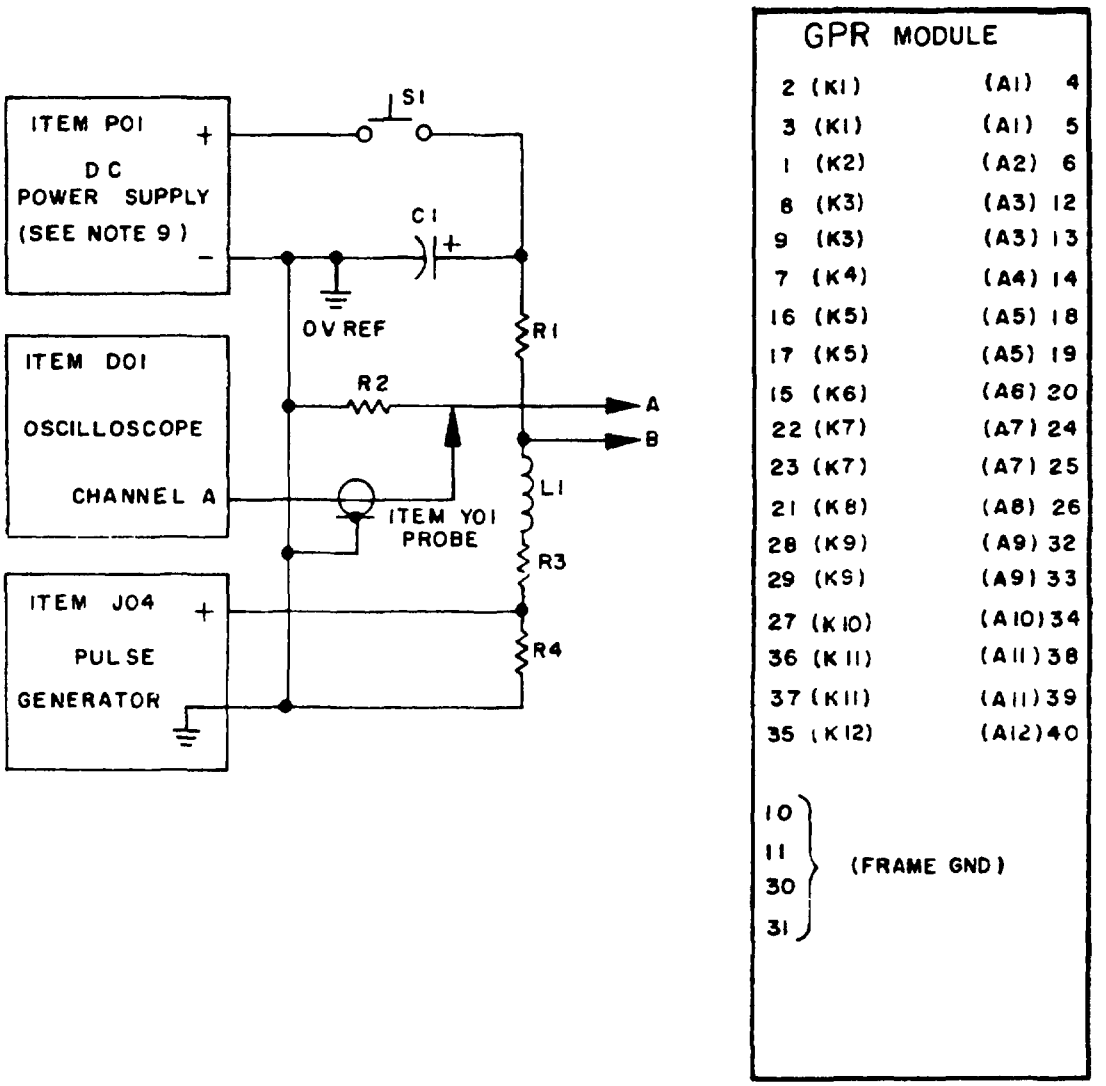
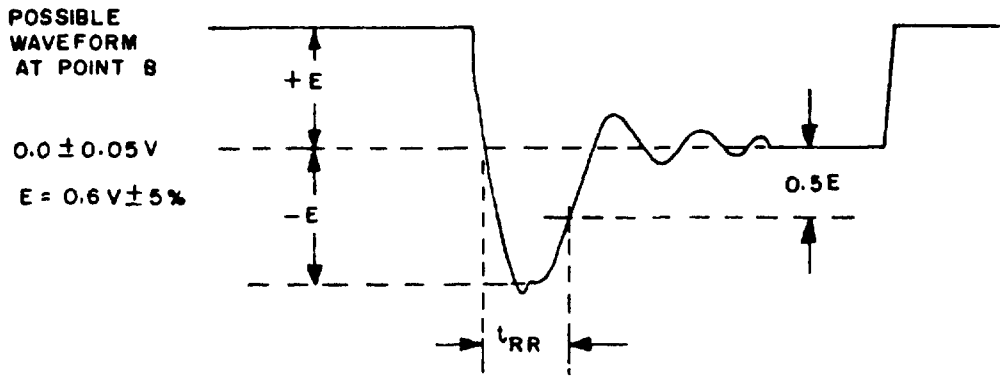


FIGURE 8. Reverse recovery time test circuit for table VIII.

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NOTES:
1. R1 = 30Ω ±5%, 50 W, non-inductive.
2. R2 = 1.0Ω ±1%, 1 W, non-inductive.
3. R3 = 11Ω ±10%, 1 W, non-inductive.
4 R4 = 100Ω ±10%, 1 W, non-inductive.
5. C1 = 500 µF -10%, +100%, 50 Vdc, electrolytic.
6. L1 = 1.0 ±0.5 µH (approximately 5 turns of number 14 or 16 wire).
7. S1 = Single pole, momentary, normally open switch.
8 Item J04 pulse parameters·
   PRP = 20 ±1 µs.
   PWA = 1.0 ±0.05 µs.
   Pulse polarity = Negative going from 0 V reference.
   $t_{TLH} = t_{THL}$ = 15 ±5 ns.
9. Tests shall be performed for each test condition as follows:

   a. Initially set item P01 and item J04 amplitude to zero.
   b. Connect points A and B per table VIII.
   c. Close S1 and adjust item P01 for a +E of 0.6 V on item D01 (-E will not appear on item D01 yet). This represents a diode forward current of 0.6 A.
   d. Adjust the item J04 amplitude for a -E of -0.6 V on item D01. This represents a diode reverse current of 0.6 A.
   e. Take reading T_{RR} from the time the negative going pulse crosses 0 V to the time it recovers to 0.5 E and does not ring below 0.5 E again. Wait 5 +5, -0 second before taking reading.

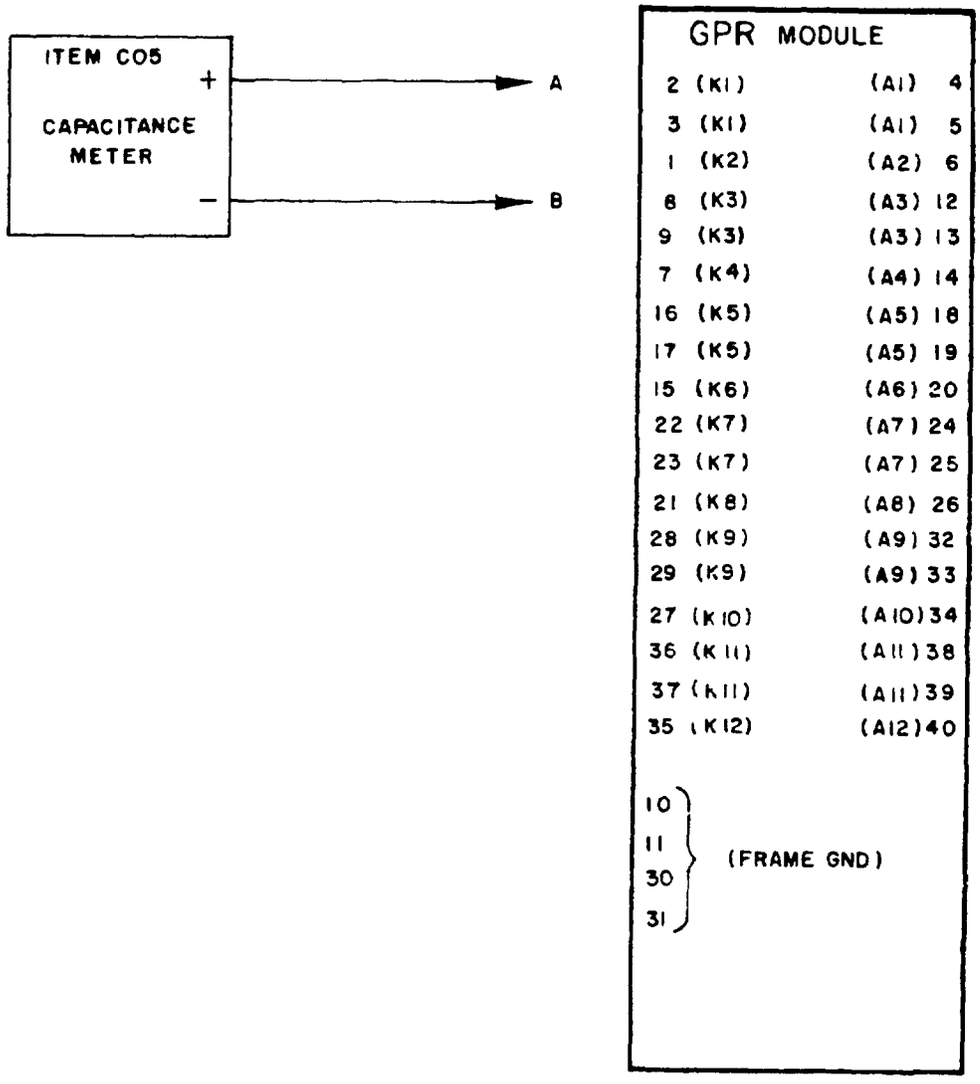
FIGURE 8 Reverse recovery time test circuit for table VIII - Continued.

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TABLE IX. Junction capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Junction capacitance	3	4	0	240	pF	9
H2	↓	1	6	↓	↓	↓	↓
H3		9	12				
H4		7	14				
H5		17	18				
H6		15	20				
H7		23	25				
H8		21	26				
H9		29	32				
H10		27	34				
H11		37	39				
H12	Junction capacitance	35	40	0	240	pF	9

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than those indicated in table IX. Stray capacitance added by other pins may affect the readings.
2 Null out the effects of the test leads prior to connecting the module.

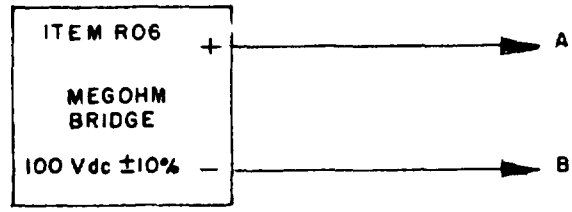
FIGURE 9 Junction capacitance test circuit for table IX

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TABLE X. Insulation resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Insulation resistance	6	2,7,8,15,16, 21,22,27,28, 35,36	10	∞	MΩ	10
J2	↓	14	1,2,8,15,16, 21,22,27,28, 35,36	↓	↓	↓	↓
J3	↓	20	1,2,7,8,16, 21,22,27,28, 35,36	↓	↓	↓	↓
J4	↓	26	1,2,7,8,15, 16,22,27,28, 35,36	↓	↓	↓	↓
J5	↓	34	1,2,7,8,15, 16,21,22,28, 35,36	↓	↓	↓	↓
J6	↓	40	1,2,7,8,15, 16,21,22,27, 28,36	↓	↓	↓	↓
J7	Insulation resistance	10,11 30,31	All other pins	10	-	MΩ	10

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

GPR MODULE

2 (K1)	(A1) 4
3 (K1)	(A1) 5
1 (K2)	(A2) 6
8 (K3)	(A3) 12
9 (K4)	(A3) 13
7 (K4)	(A4) 14
16 (K5)	(A5) 18
17 (K5)	(A5) 19
15 (K6)	(A6) 20
22 (K7)	(A7) 24
23 (K7)	(A7) 25
21 (K8)	(A8) 26
28 (K9)	(A9) 32
29 (K9)	(A9) 33
27 (K10)	(A10) 34
36 (K11)	(A11) 38
37 (K11)	(A11) 39
35 (K12)	(A12) 40

10, 11, 30, 31 } (FRAME GND)

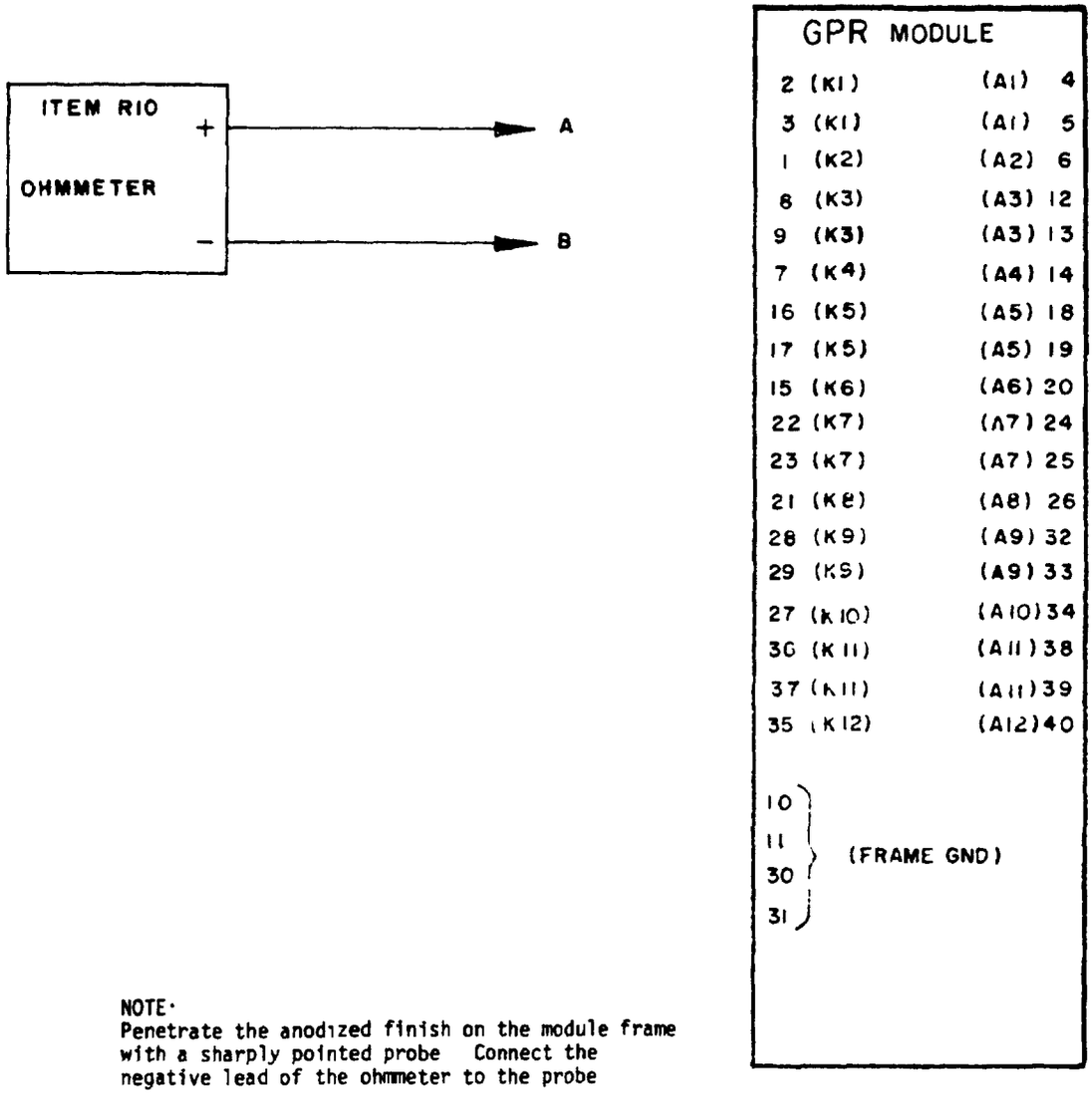
FIGURE 10 Insulation resistance test circuit for table X

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TABLE XI. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Continuity	10	Frame	0	1.0	Ω	11
K2	↓	11	↓	↓	↓	↓	↓
K3	↓	30	↓	↓	↓	↓	↓
K4	Continuity	31	Frame	0	1.0	Ω	11

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NOTE:
Penetrate the anodized finish on the module frame with a sharply pointed probe Connect the negative lead of the ohmmeter to the probe

FIGURE 11 Continuity test circuit for table XI

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TABLE XII. Peak one cycle surge current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Peak one cycle surge current	4	3	35	Note 2	A	12
L2		6	1				
L3		12	9				
L4		14	7				
L5		18	17				
L6		20	15				
L7		25	23				
L8		26	21				
L9		32	29				
L10		34	27				
L11		39	37				
L12	Peak one cycle surge current	40	35	35	Note 2	A	12

NOTES:

1. Test condition L shall be performed initially at 25°C only.
2. Each rectifier is required to meet a peak one cycle (60 Hz) surge current requirement of 35 amperes minimum. However, no current in excess of 35 amperes shall be applied to a module.

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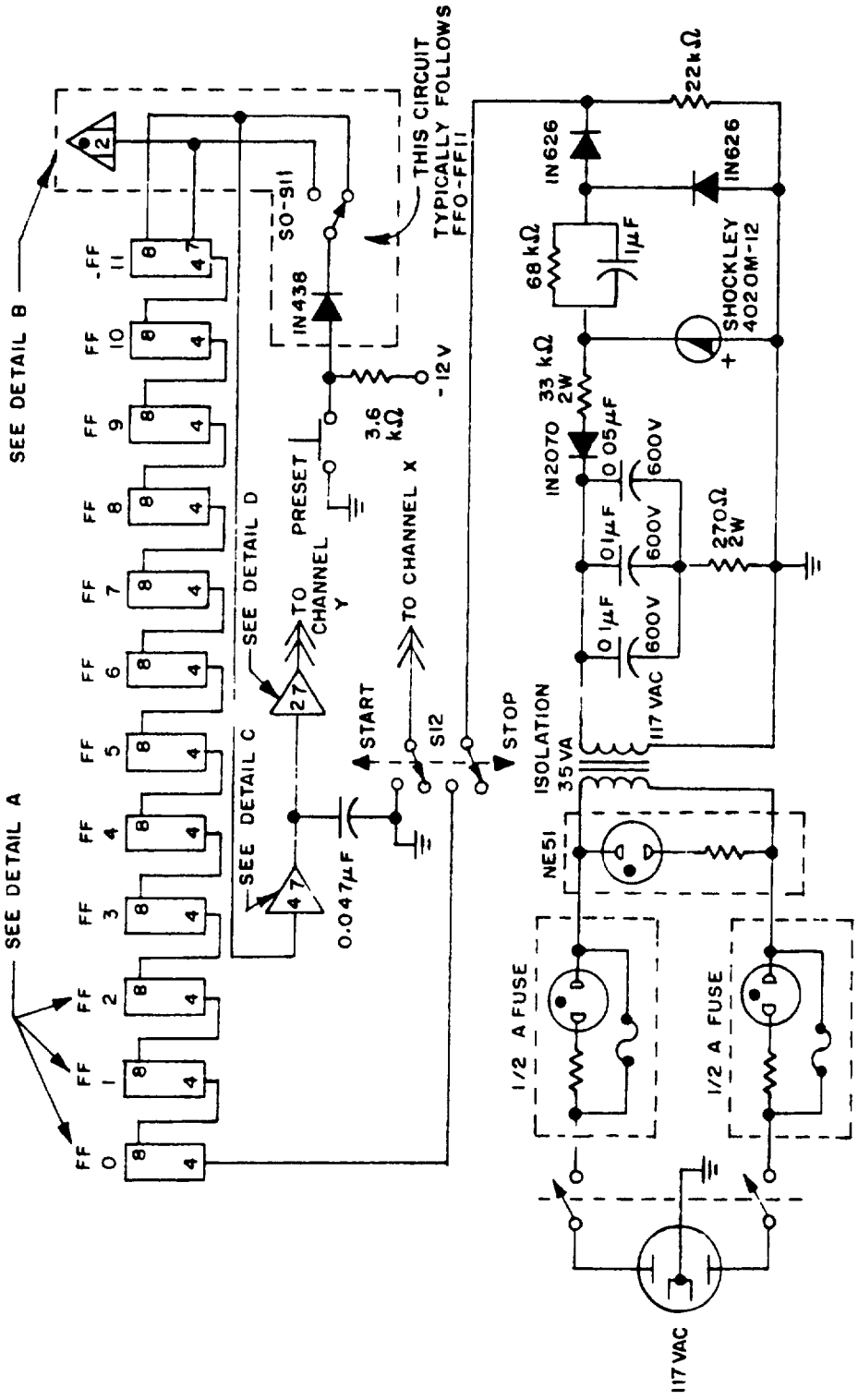


FIGURE 12. Peak one cycle surge current test circuit for table XII.

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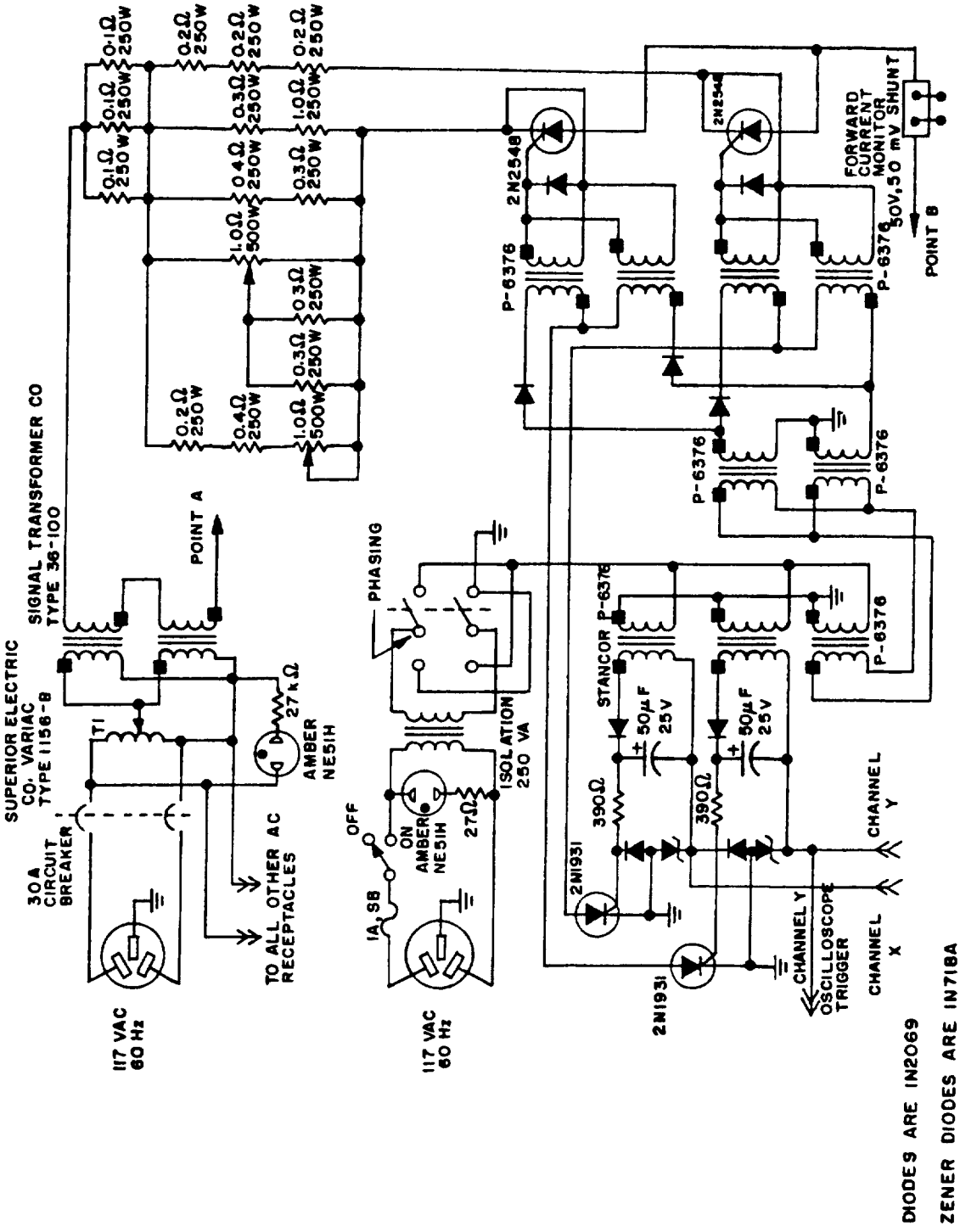


FIGURE 12. Peak one cycle surge current test circuit for table XII - Continued

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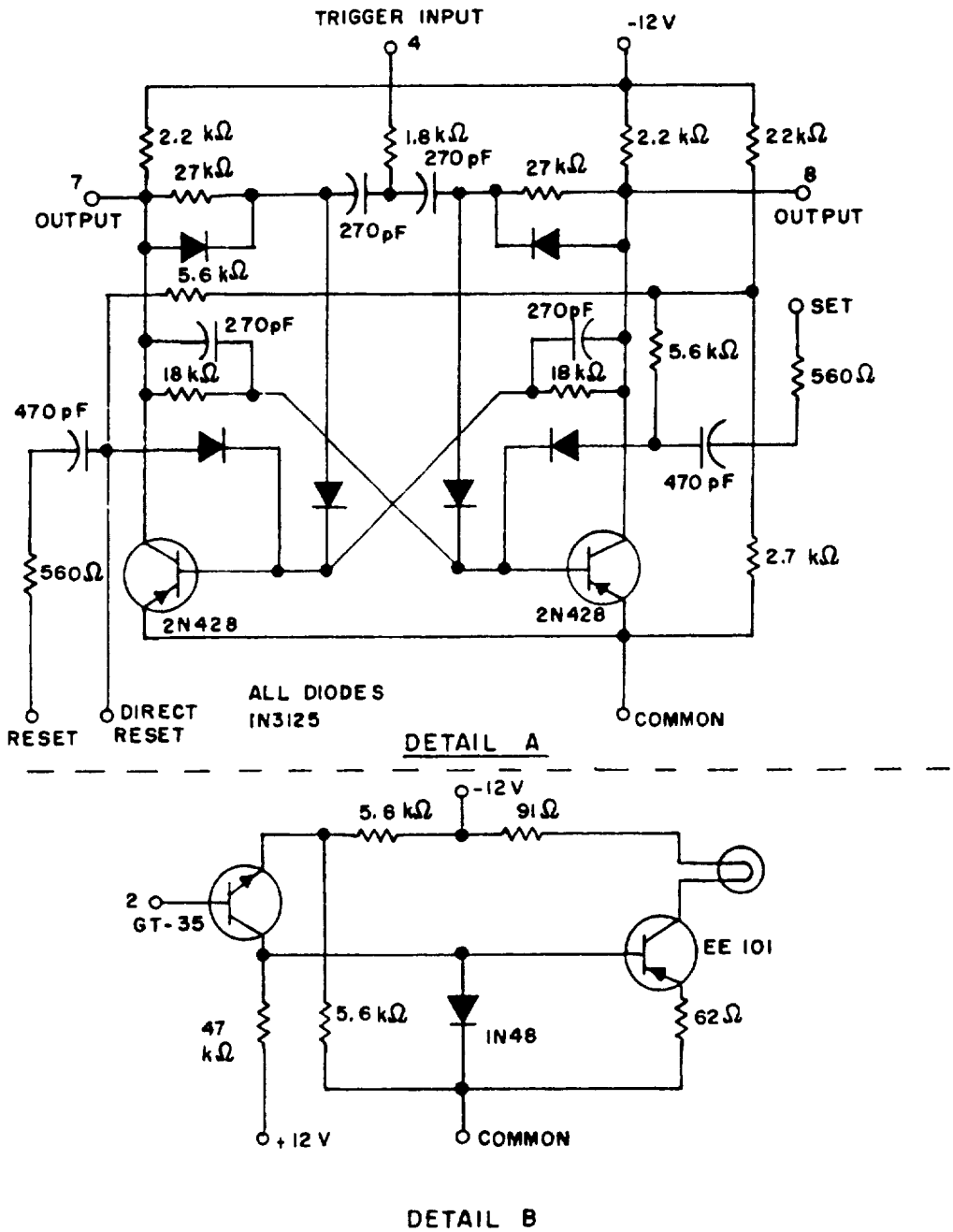
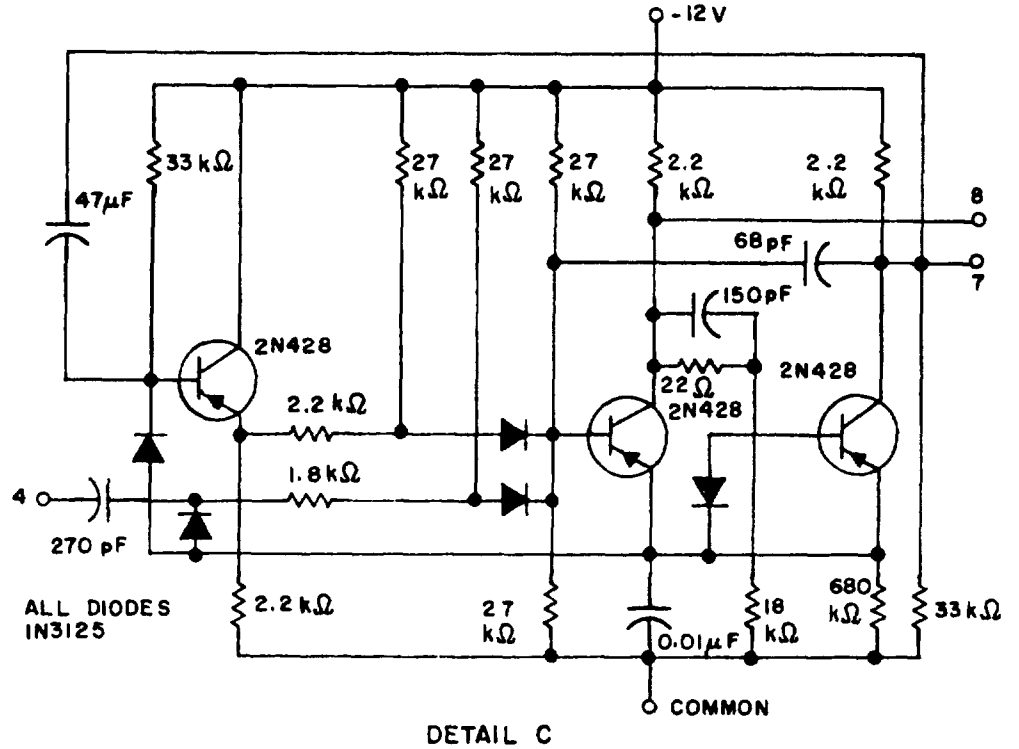
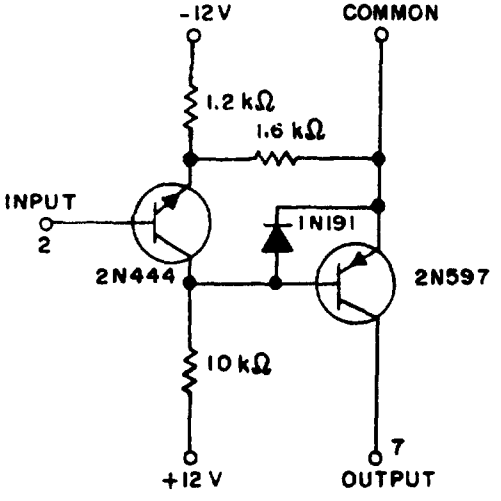


FIGURE 12 Peak one cycle surge current test circuit for table XII - Continued.

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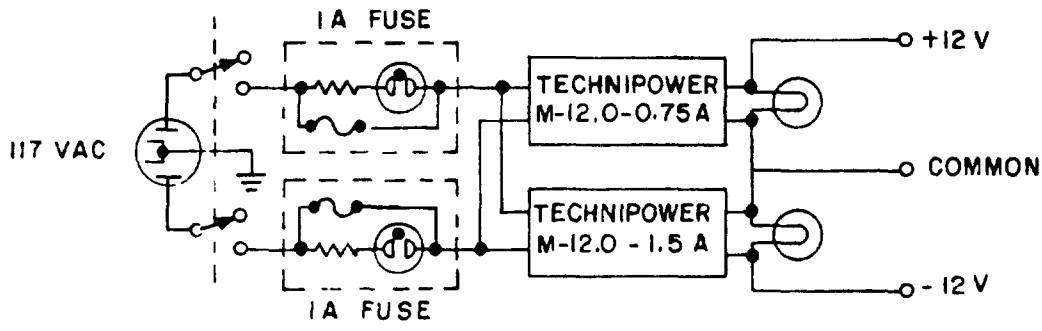
DETAIL C



DETAIL D

FIGURE 12 Peak one cycle surge current test circuit for table XII - Continued.

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NOTES
1 Switches S0 through S11 control the preset count of the binary counter When the counter fills to all 1's, channel Y will allow one cycle of 60 Hz
2 The binary counter shall be preset to a count of

FF =	0	1	2	3	4	5	6	7	8	9	10	11
Output =	1	1	1	1	0	1	1	1	1	0	0	0

This count will result in one minute or 3600 cycles of 60 Hz on channel X prior to the one cycle on channel Y.

3 Using a GPR set-up sample, with power switches "ON" and switch S12 in START, monitor current with an oscilloscope. Adjust variac T1 for a 5.50 mV indication on the oscilloscope which corresponds to 5 50 A pk. A reading of 5.50 A pk equals an average current of 1 75 A (1/2 wave I avg = 0.318 I peak)

4 The 1 Ω load potentiometers for channel Y should be adjusted for an oscilloscope indication of 35 mV which corresponds to a 35 A pk one cycle surge
5. The surge current test shall be performed one time per test condition for each module
6 All resistors are 1/2 W unless otherwise noted.
7 If the current surge test circuitry has provisions for applying a reverse voltage to the diode under test, set the reverse voltage to zero

FIGURE 12. Peak one cycle surge current test circuit for table XII - Continued

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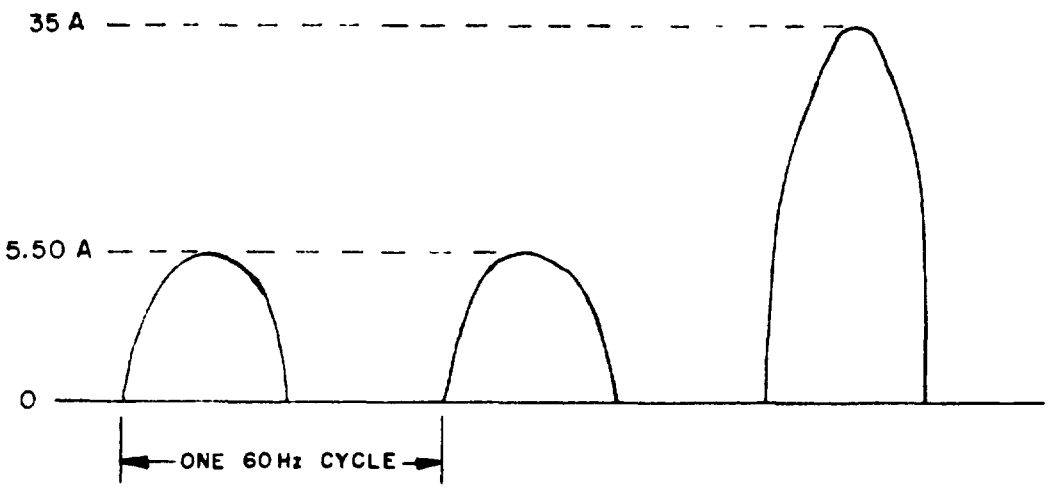


FIGURE 12. Peak one cycle surge current test circuit for table XII - Continued.

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